University of Sulaimani
College of Science
Physics Department
Forth stage
First Semester
2026-2027

Chapter 1
Vector Analysis

By
Taha Yasin Ahmed

Assistant Lecture

Learning Outcome

- **After finished this chapter, you should be able to . . .**

1. **Understand the Basics of Vector Algebra.**
2. **Compute and Interpret Dot and Cross Products.**
3. **Work with Coordinate Systems.**
4. **Apply Differential Vector Operators.**
5. **Use Integral Theorems in Field Calculations.**
6. **Visualize and Interpret Vector Fields.**

Table of Contents

01 Vector Operations

02 Vector Algebra

03 Vector Directions

04 Position Displacement

05 Del Operation

06 Second Derivatives

07 Fundamental Theorems

08 Coordinate Systems

09 Classification vector field

1.Vector Analysis

- **Scalar** is a quantity that has only magnitude, such as: time, distance, temperature, entropy, energy, mass, electric potential.

- **The vector** is a quantity that has both magnitude and direction, such as: velocity, force, displacement, electric and magnetic field intensity.

- The **physical meaning of a vector** is that it represents a quantity in physics that has both:
  1. **Magnitude** – how much (the size or amount), and
  2. **Direction** – which way it is pointing.

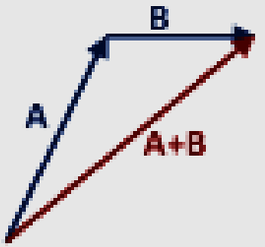
- A **vector** **tells you how much and in which direction** something is happening.

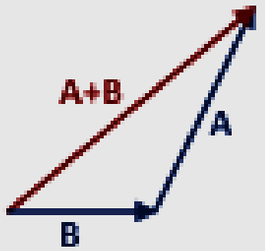
- The **use of vectors and vector analysis** can greatly simplify the mathematics used in expressing and manipulating the laws and theorems of electric and magnetic fields.

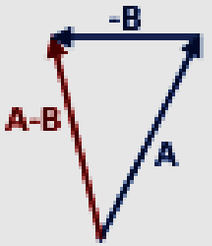
1.1 Vector Operations

1. Addition of two vectors:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$
$$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$$
$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$





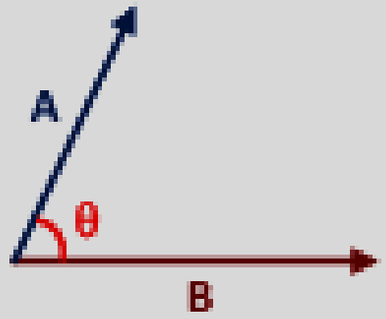


2. Multiplication by a scalar:

$$\alpha(\vec{A} + \vec{B}) = \alpha\vec{A} + \alpha\vec{B}$$

3. Dot product of two vectors (Scalar Product):

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$
$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$
$$\vec{A} \cdot \vec{A} = A^2$$
$$\vec{A} \perp \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = 0$$



$$\vec{A} \cdot \vec{B} = AB\cos(\theta)$$

1.1 Vector Operations

4. Cross product of two vectors (Vector Product):

$$\vec{A} \times \vec{B} = AB\sin(\theta)\hat{n}$$

where $\hat{n}$ is a unit vector, pointing perpendicular to the plane of $\vec{A}$ and $\vec{B}$. (the direction pointed by right-hand rule).

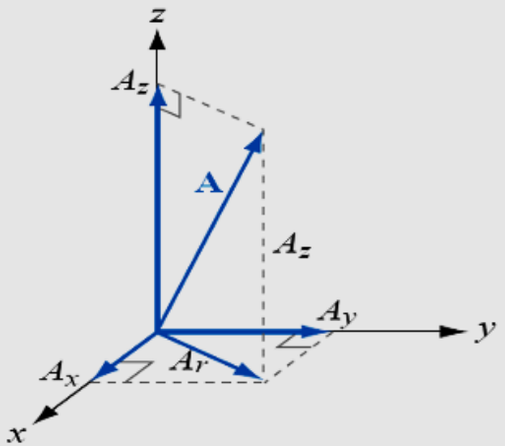
$$\vec{A} \times (\vec{B} + \vec{C}) = (\vec{A} \times \vec{B}) + (\vec{A} \times \vec{C})$$
$$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$$
$$\vec{A} \times \vec{A} = 0$$

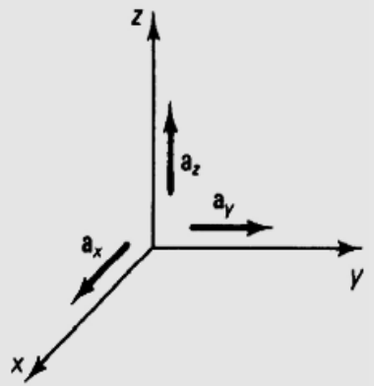
1.2 Vector Algebra

- Let $\hat{a}_x, \hat{a}_y$, and $\hat{a}_z$ be unit vectors parallel to the x, y, and z respectively. An arbitrary vector $\vec{A}$ can be expanded in term of these basis vectors:

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z$$

The numbers A_x, A_y, and A_z are called components of $\vec{A}$.





$$\vec{A} + \vec{B} = \left(A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z\right) + \left(B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z\right)$$

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{a}_x + (A_y + B_y)\hat{a}_y + (A_z + B_z)\hat{a}_z$$

$$\vec{A} \cdot \vec{B} = \left(A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z\right) \cdot \left(B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z\right)$$

$$\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z$$

$$\hat{a}_x \cdot \hat{a}_x = \hat{a}_y \cdot \hat{a}_y = \hat{a}_z \cdot \hat{a}_z = 1$$

1.2 Vector Algebra

In the case of dot product:

$$\hat{a}_x \cdot \hat{a}_x = \hat{a}_y \cdot \hat{a}_y = \hat{a}_z \cdot \hat{a}_z = 1$$

$$\hat{a}_x \cdot \hat{a}_y = \hat{a}_y \cdot \hat{a}_z = \hat{a}_z \cdot \hat{a}_x = 0$$

$$\vec{A} \cdot \vec{A} = A_x{}^2 + A_y{}^2 + A_z{}^2$$

$$|A| = \sqrt{A_x{}^2 + A_y{}^2 + A_z{}^2}$$

$$\vec{A} \cdot \hat{a}_x = A_x \; ; \; \vec{A} \cdot \hat{a}_y = A_y \; ; \; \vec{A} \cdot \hat{a}_z = A_z$$

In the case of cross product:

$$\hat{a}_x \times \hat{a}_x = \hat{a}_y \times \hat{a}_y = \hat{a}_z \times \hat{a}_z = 0$$

$$\hat{a}_x \times \hat{a}_y = \hat{a}_z$$

$$\hat{a}_y \times \hat{a}_z = \hat{a}_x$$

$$\hat{a}_z \times \hat{a}_x = \hat{a}_y$$

1.2 Vector Algebra

Therefore:

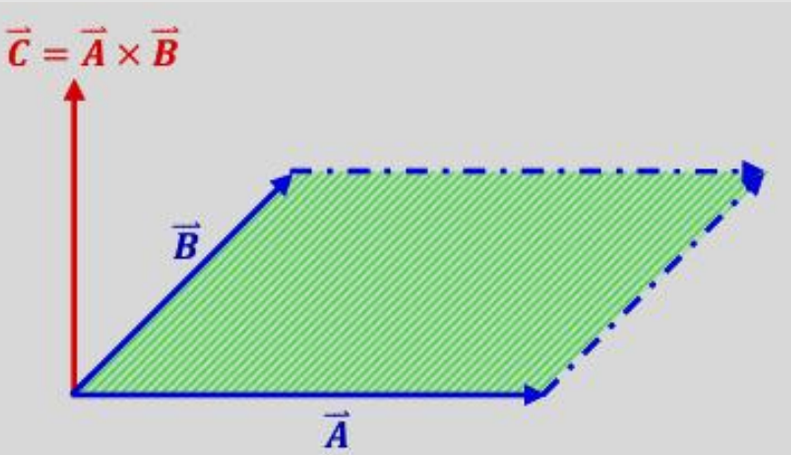
$$\vec{A} \times \vec{B} = \left(A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z\right) \times \left(B_x \hat{a}_x + B_y \hat{a}_y + B_z \hat{a}_z\right)$$

$$\vec{A} \times \vec{B} = \left(A_y B_z - A_z B_y\right)\hat{a}_x + \left(A_z B_x - A_x B_z\right)\hat{a}_y + \left(A_x B_y - A_y B_x\right)\hat{a}_z$$

This cumbersome expression can be written as a determinant:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

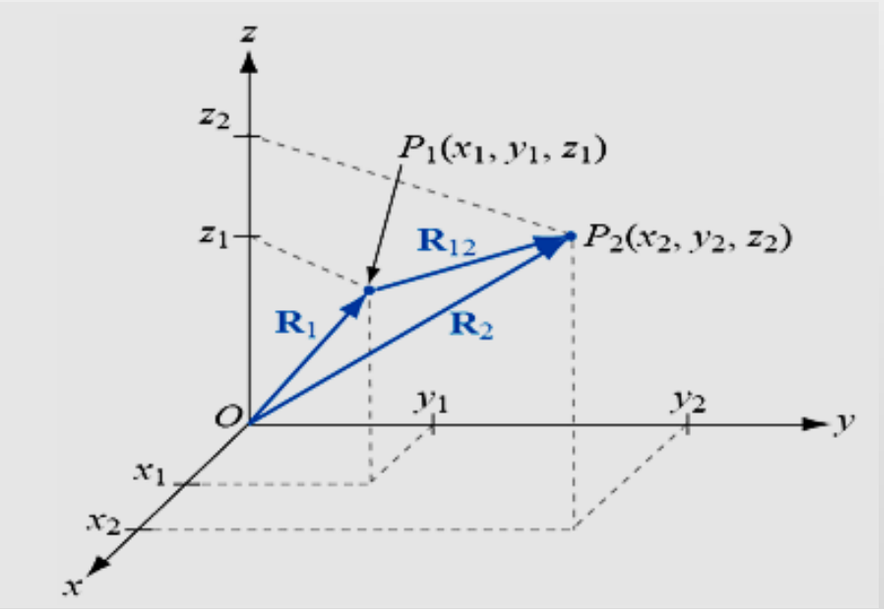
Where $\vec{C}$ is perpendicular to both $\vec{A}$ and $\vec{B}$.



1.3 Vector Directions

The position vector $\vec{R}_1$ or radius vector of point $P(x, y, z)$ is defined as the directed distance from the origin $O(0,0,0)$ to point $P(x, y, z)$ as:

$$\vec{R}_1 = \overline{OP} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$$



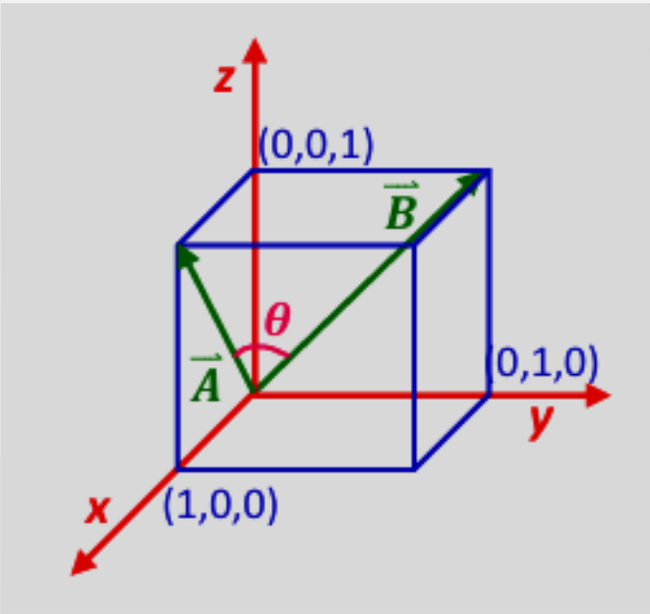
The distance vector is the displacement from one point to another, such as, $P_1(x_1, y_1, z_1)$ to $P_2(x_2, y_2, z_2)$ and is represented mathematically as:

$$\vec{R}_{12} = \overline{P_1P_2} = (x_2 - x_1)\hat{a}_x + (y_2 - y_1)\hat{a}_y + (z_2 - z_1)\hat{a}_z$$

Therefore, the position vector is a special case of displacement vector.

Example

Find the angle between the face diagonals of a cube.



Solution

$$\vec{A} = 1\hat{a}_x + 0\hat{a}_y + 1\hat{a}_z$$

$$\vec{B} = 0\hat{a}_x + 1\hat{a}_y + 1\hat{a}_z$$

$$\vec{A} \cdot \vec{B} = 0 + 0 + 1 = 1$$

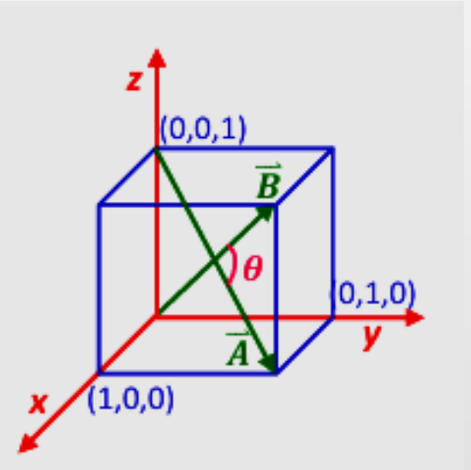
$$|A| = \sqrt{2}\,; \;\; |B| = \sqrt{2}$$

$$\vec{A} \cdot \vec{B} = AB\cos\theta \;\Rightarrow\; \cos\theta = \frac{\vec{A} \cdot \vec{B}}{AB}$$

$$\cos\theta = \frac{1}{2} \;\Rightarrow\; \theta = 60^\circ$$

Homework

Find the angle between the body diagonals of a cube.



Solution

$$\vec{A} = 1\,\hat{a}_x + 1\,\hat{a}_y - 1\,\hat{a}_z$$

$$\vec{B} = 1\,\hat{a}_x + 1\,\hat{a}_y + 1\,\hat{a}_z$$

$$\vec{A} \cdot \vec{B} = 1 + 1 - 1 = 1$$

$$|A| = \sqrt{3} \quad ; \quad |B| = \sqrt{3}$$

$$\vec{A} \cdot \vec{B} = A\,B\,cos\theta \quad \Rightarrow \quad cos\theta = \frac{\vec{A} \cdot \vec{B}}{A\,B}$$

$$cos\theta = \frac{1}{3} \quad \Rightarrow \quad \theta = 70.5288^{\circ}$$

1.4 Position Displacement

- The location of a point in three dimensions can be described by listing its Cartesian coordinates (x ,y,z). The vector to that point form the origin is called the position vector;

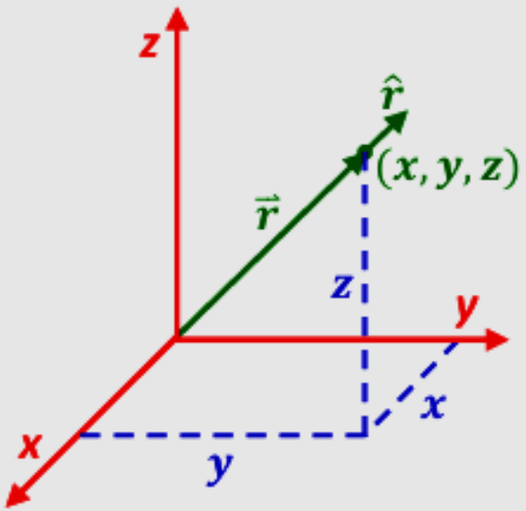
$$\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$$

Its magnitude:

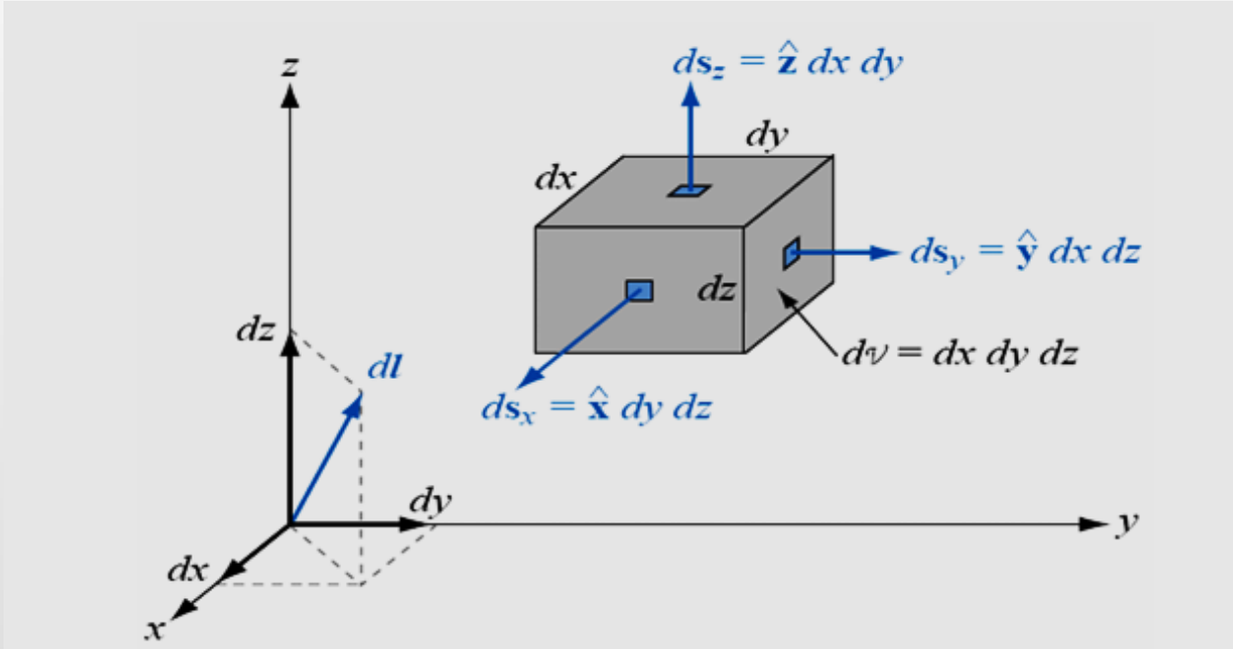
$$|r| = \sqrt{x^2 + y^2 + z^2}$$

The unit vector pointing radially outward, is

$$\hat{r} = \frac{\vec{r}}{|r|} = \frac{x\hat{a}_x + y\hat{a}_y + z\hat{a}_z}{\sqrt{x^2 + y^2 + z^2}}$$



1.4 Position Displacement



The infinitesimal displacement vector, from (x, y, z) to $(x + dx, y + dy, z + dz)$ is:

$$\overrightarrow{dl} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$$

The infinitesimal surface of Cartesian Coordinate is:

$$\vec{da} = dydz\hat{a}_x + dxdz\hat{a}_y + dxdy\hat{a}_z$$

1.5 Del Operation

1.5.1 Gradient

Suppose, we have a function of three variables, say, the temperature $T(x, y, z)$ in a room.
A theorem on partial derivative state that the derivative of T can be calculated from:

$$dT = \left(\frac{\partial T}{\partial x}\right) dx + \left(\frac{\partial T}{\partial y}\right) dy + \left(\frac{\partial T}{\partial z}\right) dz$$

This tells us how T changes when alter all three variables by the infinitesimal amount dx, dy , dz; Above equation in a dot products

$$dT = \left(\frac{\partial T}{\partial x}\hat{a}_x + \frac{\partial T}{\partial y}\hat{a}_y + \frac{\partial T}{\partial z}\hat{a}_z\right) \cdot \left(dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z\right) = \vec{\nabla}T \cdot \overrightarrow{dl}$$

Where; $\vec{\nabla}T = \frac{\partial T}{\partial x}\hat{a}_x + \frac{\partial T}{\partial y}\hat{a}_y + \frac{\partial T}{\partial z}\hat{a}_z$ → is the gradient of T

$\vec{\nabla}T$ is a vector quantity.

1.5.1 Gradient

- ✓ The gradient $\vec{\nabla}T$ points in the direction of maximum increase of the function T.
- ✓ The magnitude $|\vec{\nabla}T|$ gives the slop (rate of increase) along this maximal direction.
- ✓ If gradient vanish ($\vec{\nabla}T = 0$) at (x, y, z). this is, then, a stationary point of the function $T(x, y, z)$. It could be a maximum or a minimum points.

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z$$

is called "Del" operator.

- ✓ Del is not a vector; it is without specific meaning until we provide it with a function to act upon.

1.5.2 Divergence

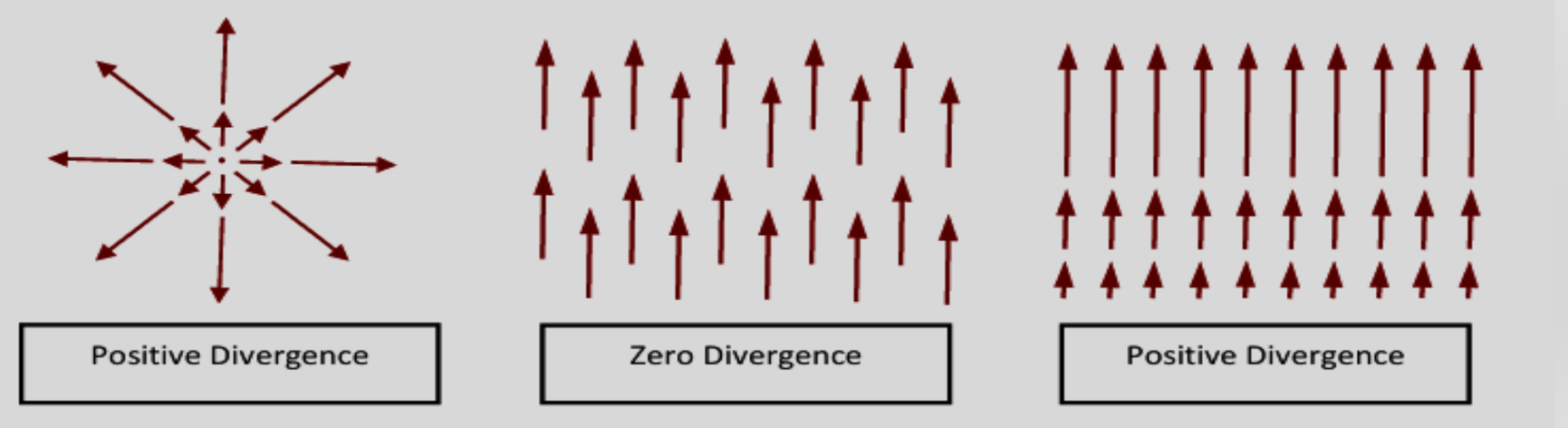
Del operator act on a vector function $\vec{V}$, via the dot product.

$$\vec{\nabla} \cdot \vec{V} = \left(\frac{\partial}{\partial x} \hat{a}_x + \frac{\partial}{\partial y} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z \right) \cdot \left(V_x \hat{a}_x + V_y \hat{a}_y + V_z \hat{a}_z \right)$$

$$\vec{\nabla} \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

The result is **scalar**.

The divergence is measure, how much the vector $\vec{V}$ spreads out



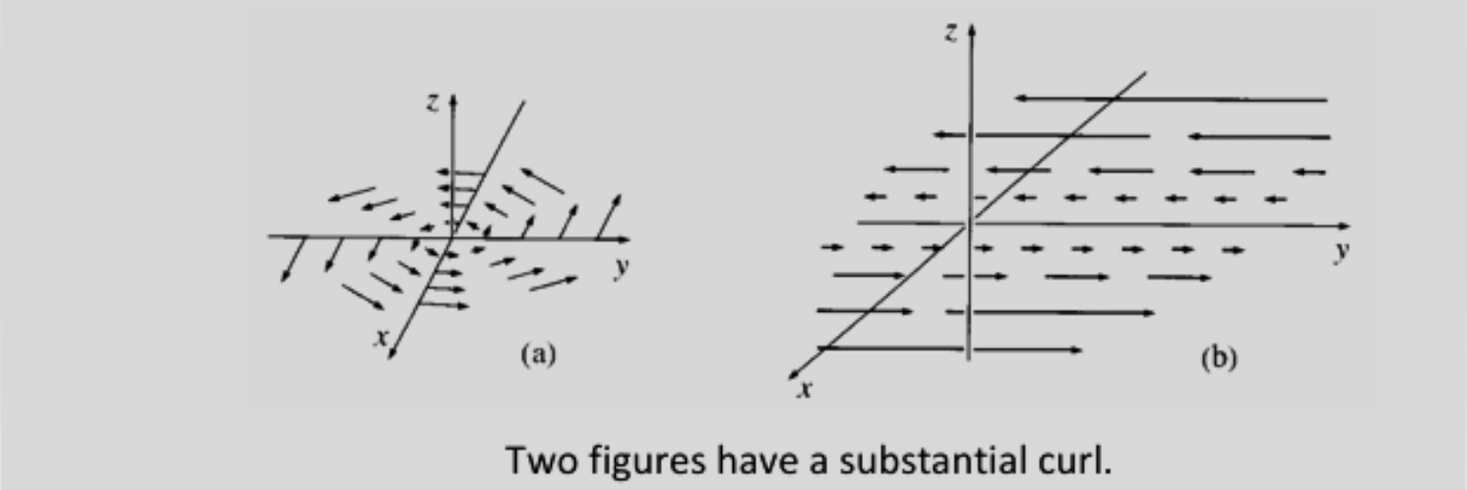
1.5.3 Curl

Del operator can act on a vector function $\vec{v}$, via the cross product:

$$\vec{\nabla} \times \vec{V} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix}$$

$$\vec{\nabla} \times \vec{V} = \hat{a}_x \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) + \hat{a}_y \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) + \hat{a}_z \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right)$$

The curl $\vec{\nabla} \times \vec{V}$ measure, how much the vector $\vec{V}$ curl around the point in question.



Two figures have a substantial curl.

1.6 Second Derivatives

- The gradient, the divergence, and curl are the only first derivative we can make with $\vec{\nabla}$; by applying $\vec{\nabla}$ twice we can construct five species of second derivatives.

✓ The **gradient** $\vec{\nabla}T$ is a **vector,** so we can take the **divergence** and **curl** of it:

1- Divergence of gradient: $\vec{\nabla} \cdot (\vec{\nabla}T)$

2- Curl of gradient: $\vec{\nabla} \times (\vec{\nabla}T)$

✓ The **divergence** $\vec{\nabla} \cdot \vec{V}$ is a **scalar**, all we can do is take its g**radient**:

3- Gradient of divergence: $\vec{\nabla}(\vec{\nabla} \cdot \vec{V})$

✓ The **curl** $\vec{\nabla} \times \vec{V}$ is a **vector**, so we can take its **divergence** and **curl**:

4- Divergence of curl: $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{V})$

5- Curl of Curl: $\vec{\nabla} \times (\vec{\nabla} \times \vec{V})$

1.6 Second Derivatives

Note that

1. $\vec{\nabla} \cdot (\vec{\nabla} T) = \nabla^2 T$, $\nabla^2 T$ is called the Laplacian of T.

Laplacian of a scalar T is a scalar.

2. The curl of a gradient is always zero: $\vec{\nabla} \times (\vec{\nabla} T) = 0$

3. $\vec{\nabla}(\vec{\nabla} \cdot \vec{V})$ is not the same as the Laplace of vector: $\nabla^2 \vec{V} = (\vec{\nabla} \cdot \vec{\nabla})\vec{V} \neq \vec{\nabla}(\vec{\nabla} \cdot \vec{V})$

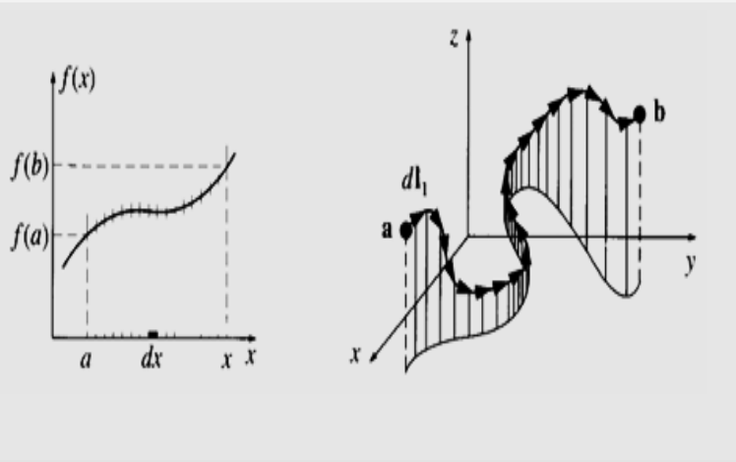
4. The Divergence of curl $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{V})$ is always zero: $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{V}) = 0$

5. From definition of $\vec{\nabla}$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{V}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{V}) - \nabla^2 \vec{V}$$

1.7 The Fundamental Theorems:

1.7.1 For gradient



Suppose we have a scalar function of three variables $T(x, y, z)$, going from a to b along the path dl, the total change in T well be:

$$\int_a^b \vec{\nabla} T \cdot \vec{dl} = T(b) - T(a)$$

Notes:
Line integrals ordinary depend on the path taken from a to b.
But the right side of above equation makes no reference to the path - only to the end points.

Corollary(1):
$\int_a^b \vec{\nabla} T \cdot \vec{dl}$ is independent of path taken from a to b.

Corollary(2):
$\oint \vec{\nabla} T \cdot \vec{dl} = 0$ since the beginning and end points are identical, and hence $T(b) - T(a) = 0$.

1.7.2 For Divergences

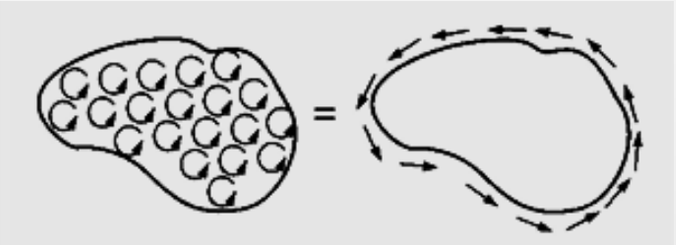
The fundamental theorem for divergences state that:

$$\int_v (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_s \vec{V} \cdot \overrightarrow{da}$$

This theorem has at least three special names:
Gauss's theorem , **Green's theorem** , or, simply, the **Divergence theorem**.

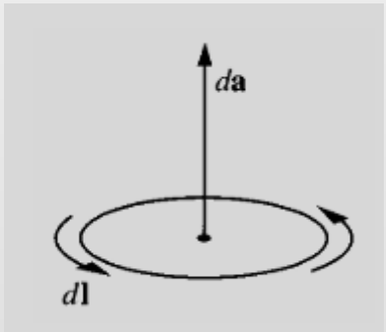
It say that the integral of a derivative (in this case the divergence) over a region (in this case a volume) is equal to the value of the function at the boundary (in this case the surface) that bounds the volume.

1.7.3 For Curls



- The fundamental theorem of Curls, which goes by the special name of **Stokes' theorem** state that:

$$\int_s (\vec{\nabla} \times \vec{V}) \cdot \overrightarrow{da} = \oint \vec{\nabla} \cdot \overrightarrow{dl}$$



As always, the integral of a derivative (here, the curl) over a region (here, a patch of surface) is equal to value of the function at the boundary(here, the perimeter of the patch).

- If your finger point in the direction of the line integral, then your thumb fixed the direction of $\overrightarrow{da}$.

Corollary(1):
$\int . (\vec{\nabla} \times \vec{V}) \cdot \overrightarrow{da}$ depends only on the boundary line, not on the particular surface used.

Corollary(2):
$\oint (\vec{\nabla} \times \vec{V}) \cdot \overrightarrow{da} = 0$ for any closed surface.

1.8 Orthogonal Coordinate Systems:

The most standard orthogonal coordinate systems that are commonly used are:

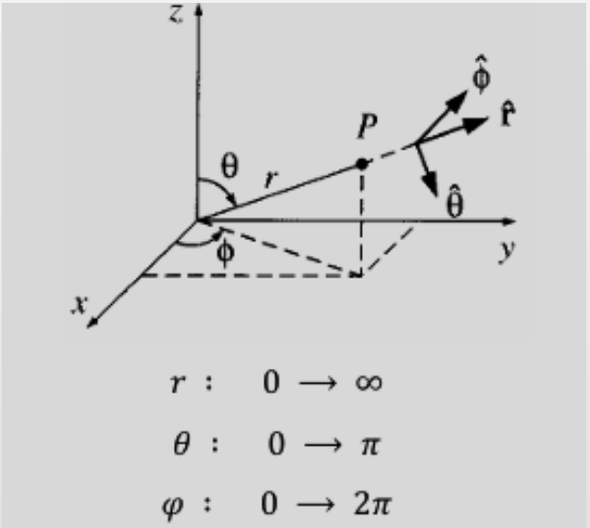
(1) Cartesian (or Rectangular) Coordinate System (x, y, z)

(2) Spherical Coordinate System (r, θ, φ)

(3) Cylindrical Coordinate System (ρ, φ, z)

1.8 Orthogonal Coordinate Systems:

1.8.1 Spherical Polar Coordinates



The spherical plot coordinates (r, θ, φ) of a point P are defined in Fig;

r: is the distance from the origin (the magnitude of the position vector).
θ : the angle down from the z-axis, is called the polar angle.
φ : the angle around from the x-axis, is the azimuthal angle.

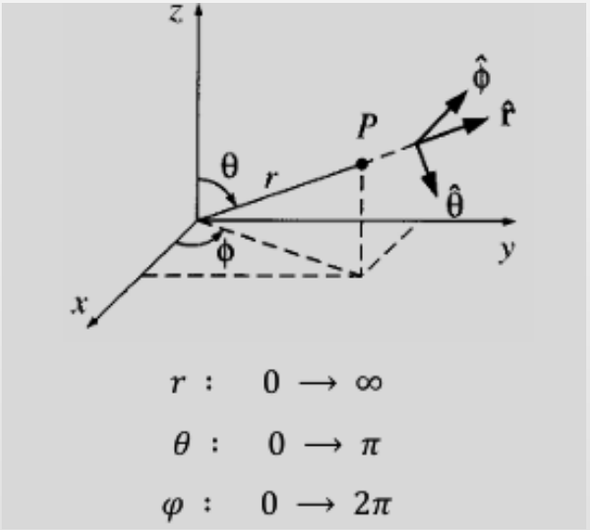
Their relation to Cartesian coordinates (x, y, z) can read from the figure.

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$

1.8.1 Spherical Polar Coordinates

✓ Any vector $\vec{A}$ can be expressed as:

$$\vec{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_\varphi \hat{a}_\varphi$$



$A_r, A_\theta \& A_\varphi$ are the radial, polar & azimuthal components of $\vec{A}$.
$\hat{a}_r, \hat{a}_\theta \& \hat{a}_\varphi$ are unit vectors, pointing in the direction of increase of the corresponding coordinates.

✓ In terms of the Cartesian unit vectors:

$$\hat{a}_r = \sin\theta \cos\varphi \, \hat{a}_x + \sin\theta \sin\varphi \, \hat{a}_y + \cos\theta \, \hat{a}_z$$

$$\hat{a}_\theta = \cos\theta \cos\varphi \, \hat{a}_x + \cos\theta \sin\varphi \, \hat{a}_y - \sin\theta \, \hat{a}_z$$

$$\hat{a}_\varphi = -\sin\varphi \, \hat{a}_x + \cos\varphi \, \hat{a}_y$$

1.8.1 Spherical Polar Coordinates

Check: $\hat{a}_r . \hat{a}_r = \hat{a}_\theta . \hat{a}_\theta = \hat{a}_\varphi \cdot \hat{a}_\varphi = 1$

$$\hat{a}_r \cdot \hat{a}_\theta = \hat{a}_\theta \cdot \hat{a}_\varphi = \hat{a}_\varphi \cdot \hat{a}_r = 0$$

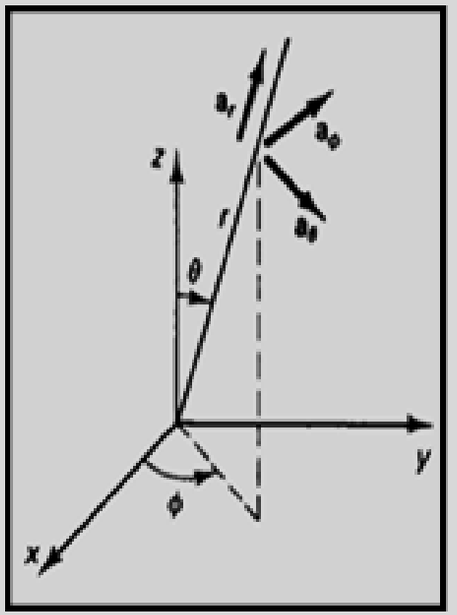
$$\hat{a}_r \times \hat{a}_\theta = \hat{a}_\varphi \; ; \; \hat{a}_\theta \times \hat{a}_\varphi = \hat{a}_r \; ; \; \hat{a}_\varphi \times \hat{a}_r = \hat{a}_\theta$$

·	$\hat{a}_x$	$\hat{a}_y$	$\hat{a}_z$
$\hat{a}_r$	$\sin\theta\cos\varphi$	$\sin\theta\sin\varphi$	$\cos\theta$
$\hat{a}_\theta$	$\cos\theta\cos\varphi$	$\cos\theta\sin\varphi$	$-\sin\theta$
$\hat{a}_\varphi$	$-\sin\varphi$	$\cos\varphi$	0

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$$

$$\varphi = \tan^{-1}\left(\frac{y}{x}\right)$$



1.8.1 Spherical Polar Coordinates

An infinitesimal displacement in the $\hat{a}_r$ direction is:

$$dl_r = dr$$

An infinitesimal element of length in the $\hat{a}_\theta$ direction is:

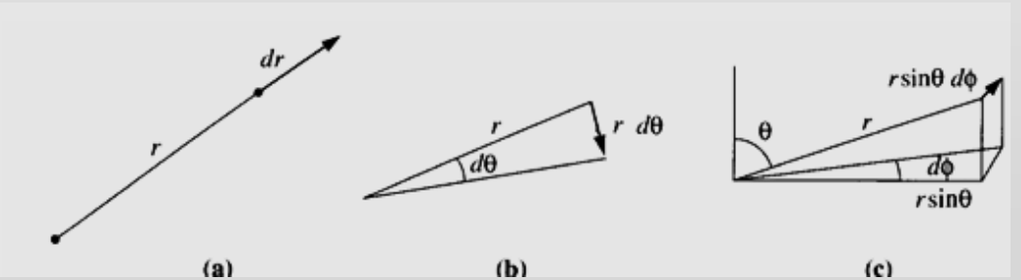
$$dl_\theta = rd\theta$$

An infinitesimal element of length in the $\hat{a}_\varphi$ direction is:

$$dl_\varphi = r\sin\theta d\varphi$$

✓ Thus, the general infinitesimal **displacement** dl is:

$$\overrightarrow{dl} = dr\hat{a}_r + rd\theta\hat{a}_\theta + r\sin\theta d\varphi\hat{a}_\varphi$$



1.8.1 Spherical Polar Coordinates

✓ The infinitesimal **volume element** $d\tau$, in spherical coordinates, is:

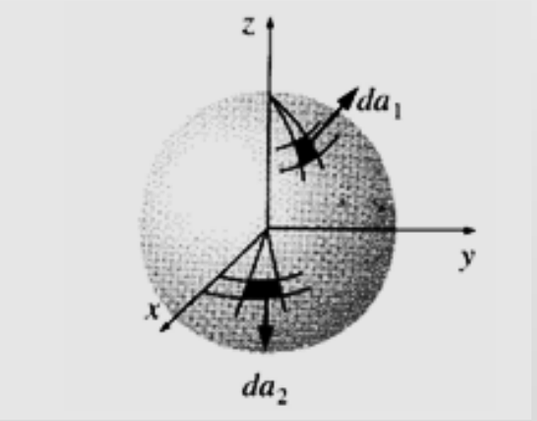
$$d\tau = dl_r dl_\theta dl_\varphi$$

$$d\tau = r^2 \sin\theta dr d\theta d\varphi$$

✓ For surface elements $\overrightarrow{da}$:

1- If r is constant:

$$da_1 = dl_\theta dl_\varphi \hat{a}_r = r^2 \sin\theta d\theta d\varphi \hat{a}_r$$



2- If θ is constant:

$$da_2 = dl_r dl_\varphi \hat{a}_\theta = r \sin\theta dr d\varphi \hat{a}_\theta$$

3- If φ is constant:

$$da_3 = dl_r dl_\theta \hat{a}_\varphi = r dr d\theta \hat{a}_\varphi$$

In general:

$$\overrightarrow{da} = r^2 \sin\theta d\theta d\varphi \hat{a}_r + r \sin\theta dr d\varphi \hat{a}_\theta + r dr d\theta \hat{a}_\varphi$$

1.8.1 Spherical Polar Coordinates

✓ The infinitesimal **displacement** dl is:

$$\overrightarrow{dl} = dr\hat{a}_r + rd\theta\hat{a}_\theta + r\sin\theta d\varphi\hat{a}_\varphi$$

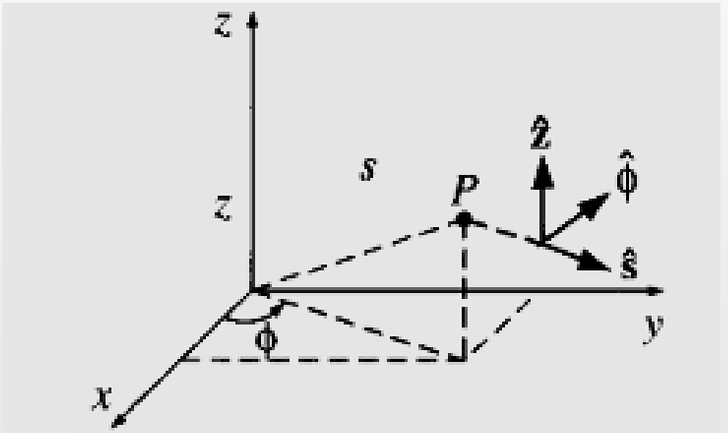
✓ The infinitesimal **volume element** $d\tau$ is:

$$d\tau = r^2\sin\theta drd\theta d\varphi$$

✓ The infinitesimal **surface area** $\overrightarrow{da}$ is :

$$\overrightarrow{da} = r^2\sin\theta d\theta d\varphi\hat{a}_r + r\sin\theta drd\varphi\hat{a}_\theta + rdrd\theta\hat{a}_\varphi$$

1.8.2 Cylindrical Coordinates



The cylindrical coordinates (ρ, φ, z) of a point P are defined in Fig.

$$\begin{aligned} \rho&: \quad 0 \quad \to \infty \\ \varphi&: \ 0 \quad \to 2\pi \\ z&: -\infty \quad \to +\infty \end{aligned}$$

r : is the distance from the z-axis.

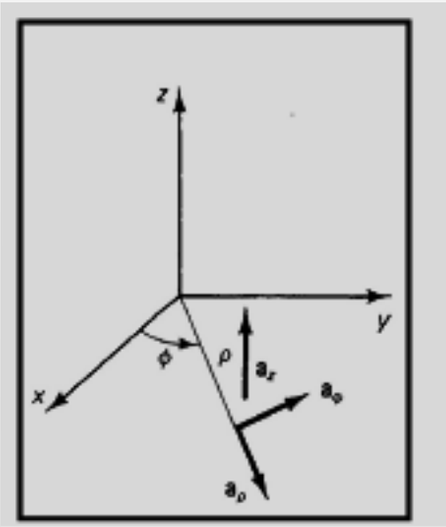
Their relation to Cartesian coordinates (x, y, z) is:

$$x = \rho\cos\varphi$$
$$y = \rho\sin\varphi$$
$$z = z$$
$$\rho = \sqrt{x^2 + y^2}$$
$$\varphi = \tan^{-1}\left(\frac{y}{x}\right)$$

1.8.2 Cylindrical Coordinates

Any vector $\vec{A}$ can be expressed as:

$$\vec{A} = A_\rho \hat{a}_\rho + A_\varphi \hat{a}_\varphi + A_z \hat{a}_z$$



$\hat{a}_\rho, \hat{a}_\varphi \& \hat{a}_z$ are unit vectors, pointing in the direction of increase of the corresponding coordinates. In terms of the Cartesian unit vectors:

$$\hat{a}_\rho = \cos\varphi \hat{a}_x + \sin\varphi \hat{a}_y$$
$$\hat{a}_\varphi = -\sin\varphi \hat{a}_x + \cos\varphi \hat{a}_y$$
$$\hat{a}_z = \hat{a}_z$$

·	$\hat{a}_x$	$\hat{a}_y$	$\hat{a}_z$
$\hat{a}_\rho$	$\cos\varphi$	$\sin\varphi$	0
$\hat{a}_\varphi$	$-\sin\varphi$	$\cos\varphi$	0
$\hat{a}_z$	0	0	1

1.8.2 Cylindrical Coordinates

The infinitesimal displacements in cylindrical coordinates are:

$$dl_\rho = d\rho$$
$$dl_\varphi = \rho d\varphi$$
$$dl_z = dz$$

✓ Thus, the general **infinitesimal displacement** dl is:

$$\overrightarrow{dl} = d\rho\hat{a}_\rho + \rho d\varphi\hat{a}_\varphi + dz\hat{a}_z$$

✓ The infinitesimal **surface elements** $\overrightarrow{da}$, in cylindrical coordinates, is:

$$\overrightarrow{da} = \rho d\varphi dz\hat{a}_\rho + d\rho dz\hat{a}_\varphi + \rho d\rho d\varphi\hat{a}_z$$

✓ The infinitesimal **volume element** $d\tau$, in cylindrical coordinates, is:

$$d\tau = \rho d\rho d\varphi dz$$

1.9 Classification of Vector Field

A vector field is uniquely characterized by its divergence and curl. All vector fields can be classified in terms of their vanishing or non-vanishing divergence or curl as follows:

(1) A vector field $\vec{A}$ is said to be **Solenoid** (**or divergence less**) if $(\vec{\nabla} \cdot \vec{A} = 0)$ [such field has neither source nor sink of flux].

Examples of such fields are (magnetic field, conduction current density under steady state condition). The Solenoid field $\vec{A}$ can always be expressed in terms of another vector such as $\vec{F}$:

$$\vec{\nabla} \cdot \vec{A} = 0 \Rightarrow \vec{A} = \vec{\nabla} \times \vec{F} \Rightarrow \vec{\nabla} \cdot \vec{\nabla} \times \vec{F} = 0$$

(2) A vector field $\vec{A}$ is said to be **irrotational** (or **conservative**) if $\vec{\nabla} \times \vec{A} = 0$, examples of such fields are [Electrostatic field and gravitational field].

Thus, an irritation field $\vec{A}$ can always be expressed in terms of a gradient of scalar field V, that is:

$$\vec{\nabla} \times \vec{A} = 0 \Rightarrow \vec{A} = \vec{\nabla} V \Rightarrow \vec{\nabla} \times \vec{\nabla} V = 0$$

1.9 Classification of Vector Field

(3) The field is said to be divergence if $\vec{\nabla} \cdot \vec{A} \neq 0$.

(4) The field is said to be Rotational (or non-conservative) if $\vec{\nabla} \times \vec{A} \neq 0$.

(5) When $\vec{\nabla}^2 V = 0$, the field is said to be harmonic.

(6) When the field is independent on position the field is said to be uniform field.

(7) When the field is independent on time the field is said to be constant field.


Thank you!
End